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教育经历

2018.09 - 2022.07
北京大学 流体力学 博士 (导师: 陈十一院士)
2014.09 - 2018.07
北京大学 理论与应用力学 本科

研究方向

湍流, 传热传质, 多相流, 智能科学计算

科研项目

国自然青年科学基金, 壁湍流中展向旋转与法向分层效应的相似性研究, 30 万元, 2024.01-2026.12, 主持
国自然区域创新发展联合基金集成项目课题, 基于人工智能的多物理场仿真计算理论基础、算法与应用-课题 5, 150 万元, 2026.01-2029.12, 主持
国家科技重大专项子课题, 锻造工艺宏/微观并行耦合仿真计算算法研究, 40 万元, 2024.10-2027.09, 主持
宁波市甬江人才工程青年创新人才项目, 旋转壁湍流控制优化的数值模拟研究, 100 万元, 2025.01-2029.12, 主持
宁波市重点研发计划项目, 基于人工智能的复杂流动建模与快速仿真技术与软件平台研发, 100 万元, 2023.04-2026.04, 主持
宁波市顶尖人才科技项目, 智能制造中的流动分析建模与仿真软件开发, 10000 万元, 2023.07-2028.06, 第三完成人
企业研发课题, 新型固定床反应器设计与数值模拟, 15 万元, 2024.10-2025.04, 主持

奖励荣誉

浙江省高层次人才, 2025
宁波市领军人才, 2025
宁波市甬江人才工程青年创新人才, 2024
北京市优秀博士学位论文, 2024
北京大学优秀毕业生, 2022

学术论文 (通讯作者以*标记)

Feixiong Rao, Shengqi Zhang*, (2026). Unified scaling laws in porous-medium vertical natural convection. *Journal of Fluid Mechanics*, 1028: A17.
Yang Chen, Pingang Wei, Shengqi Zhang, Zhanming Wan, Peng Zhang, Kaixuan Zhang, Peng Wang, Wei Liu*, (2026). Pattern selection between two-dimensional Rayleigh-Bénard convection and electroconvection near electrochemical surfaces. *Journal of Fluid Mechanics*, 1028: A10.

- Lei Du, Shengqi Zhang*, Ruili Zhang, Shibin Zhang, (2026). Interpolation-based parametric reduced-order models with dynamic mode decomposition. *Journal of Computational Physics*, 544: 114436.
- Zhenze Zhu, Shengqi Zhang*, Lifeng Xiao, (2025). Numerical study of methane dry reforming in an electromagnetic induction heating reactor. *Applied Thermal Engineering*, 280(5): 128514.
- Feixiong Rao, Shengqi Zhang*, (2025). Coupled modelling of turbulent forced thermal convection in anisotropic porous media. *Journal of Fluid Mechanics*, 1017: A31.
- Lei Du, Shengqi Zhang*, Yongchao Yang, (2025). Interpolation-based parametric reduced-order models via Galerkin projection and dynamic mode decomposition. *Physics of Fluids*, 37(8): 087169.
- Feixiong Rao, Shengqi Zhang*, (2025). Heat-fluid-solid coupling model for turbulent forced convection within porous media. *Physical Review Fluids*, 10(6): 064502.
- Jin Hu, Shengqi Zhang, Zhenhua Xia*, (2025). An efficient modified enthalpy method for phase-change melting problems with dual time-steps. *International Journal of Heat and Mass Transfer*, 239: 126544.
- Kaizhen Shi, Shengqi Zhang*, (2025). Turbulent transport in annular Poiseuille flow with axial rotation. *Physics of Fluids*, 37(3): 035116.
- Jun Lai*, Yiming Qi, Shengqi Zhang, Lian-Ping Wang*, (2025). Analytical solutions of layered Poiseuille flows in the diffuse interface model. *Physics of Fluids*, 37(2), 023612.
- Feixiong Rao*, Shengqi Zhang, (2025). The in-depth analysis and validation of momentum dispersion modeling based on direct numerical simulation. *Physics of Fluids*, 37(1): 015142.
- Qifan Wang, Shengqi Zhang, Taipeng Guo, Wenxuan Sha, Keyang Li, Zhentao Liu*, (2024). Enhancing plate-fin heat exchanger hydraulic thermal performance through air-side fin optimization based on pseudo-3D topology optimization. *Applied Thermal Engineering*, 252: 123642.
- Jin Hu, Shengqi Zhang, Zhishun Yang, Lihua Chen, Zhenhua Xia*, (2024). Numerical investigations of melting process with sidewall heating under various control parameters. *International Journal of Heat and Mass Transfer*, 227: 125529.
- Jin Hu, Shengqi Zhang, Zhenhua Xia*, (2024). Flow reversal and multiple states in turbulent Rayleigh-Bénard convection with partially isothermal plates. *Journal of Fluid Mechanics*, 987: A9.
- Shengqi Zhang*, Chao Sun, (2024). Twin forces: similarity between rotation and stratification effects on wall turbulence. *Journal of Fluid Mechanics*, 979: A45.
- Shengqi Zhang, Zhenhua Xia*, Shiyi Chen, (2024). Enhancing large-scale motions and turbulent transport in rotating plane Poiseuille flow. *Journal of Fluid Mechanics*, 979: A43.
- Qifan Wang, Shengqi Zhang, Yu Zhang, Jiahong Fu, Zhentao Liu*, (2023). Enhancing performance of nanofluid mini-channel heat sinks through machine learning and multi-objective optimization of operating parameters. *International Journal of Heat and Mass Transfer*, 210: 124204.
- Jun Lai, Tao Chen, Shengqi Zhang, Zuoli Xiao, Shiyi Chen, Lian-Ping Wang*, (2022). A systematic study of a droplet breakup process in decaying homogeneous isotropic turbulence using a mesoscopic simulation approach. *Journal of Turbulence*, 23(11-12): 567-614.
- Jin Hu, Shengqi Zhang, Zhenhua Xia*, (2022). Flow field and heat transfer properties in turbulent convection with local heating condition at the bottom wall. *Acta Aerodynamica Sinica*, 40(2): 208-214.
- Shengqi Zhang, Zhenhua Xia*, Shiyi Chen, (2022). Flow structures in spanwise rotating plane Poiseuille flow based on thermal analogy. *Journal of Fluid Mechanics*, 933: A24.
- Shengqi Zhang, Zhenhua Xia*, Shiyi Chen*, (2022). Perturbation analysis of baroclinic torque in low-Mach-number flows. *Journal of Fluid Mechanics*, 930: A4.
- Shengqi Zhang, Zhenhua Xia*, (2022). Capturing the Baroclinic Effect in non-Boussinesq Gravity Currents. *Theoretical and Applied Mechanics Letters*, 12(1): 100313.

Shengqi Zhang, Xin Chen, Zhenhua Xia*, Heng-Dong Xi, Quan Zhou, Shiyi Chen*, (2021). Stabilizing/destabilizing the large-scale circulation in turbulent Rayleigh–Bénard convection with sidewall temperature control. *Journal of Fluid Mechanics*, 915: A14.

Shengqi Zhang, Zhenhua Xia*, Shiyi Chen, (2021). A CFD-aided Galerkin Method for Global Linear Instability Analysis. *Communications in Computational Physics*, 29(1): 128-147.

Shengqi Zhang, Zhenhua Xia*, Quan Zhou, Shiyi Chen*, (2020). Controlling flow reversal in two-dimensional Rayleigh–Bénard convection. *Journal of Fluid Mechanics*, 891: R4.

Shengqi Zhang, Zhenhua Xia*, Yipeng Shi, Shiyi Chen*, (2019). A two-dimensional-three-component model for spanwise rotating plane Poiseuille flow. *Journal of Fluid Mechanics*, 880: 478-496.

Liren Li, Yipeng Shi, Shengqi Zhang, Lian-Ping Wang, Zhenhua Xia*, (2019). On the comparison between lattice boltzmann methods and spectral methods for DNS of incompressible turbulent channel flows on small domain size. *Advances in Applied Mathematics and Mechanics*, 11(3): 598-607.

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张蕊丽, 章盛祺, (2026). 保路径曲边 P2 网格自适应生成方法、装置、设备及介质. 发明专利, ZL202512003381.X.

赖俊, 章盛祺, (2026). 一种两相流动传热的数值模拟方法及系统. 发明专利, ZL202411602864.0.

杜磊, 章盛祺, (2025). DMD 参数化降阶仿真软件 V1.0. 软件著作权, 2025SR2160817.

杜磊, 章盛祺, (2025). DMD-Galerkin 参数化降阶仿真软件 V1.0. 软件著作权, 2025SR1740896.

张蕊丽, 章盛祺, (2025). 基于黎曼度量的二维曲边网格生成与自适应软件 V1.0. 软件著作权, 2025SR1719524.

邀请报告

“Validity of buoyancy approximations in low-Mach-number flows”, *China-Japan-Korea Symposium on Thermal and Fluid Engineering*, 2026 年 1 月, 杭州.

“剪切流中密度分层与展向旋转效应的相似性”, 大西南应用数学和力学研讨会, 2025 年 8 月, 凉山.

“剪切流中密度分层与展向旋转效应的相似性”, 全国湍流与流动稳定性专题研讨会, 2025 年 4 月, 南昌.

“壁湍流中展向旋转与法向分层效应的相似性”, 湍流结构的生成演化与作用机理重大研究计划年度交流会, 2024 年 1 月, 陵水.

“Statistics and parameter reduction of rotating stratified wall turbulence”, *Advanced Summer Program on Turbulence, Noise and CFD Methods*, 2023 年 8 月, 香港.

“壁湍流中展向旋转与法向分层效应的相似性”, 全国湍流与流动稳定性专题研讨会, 2023 年 4 月, 宁波.

“低马赫数流动中斜压矩的摄动展开”, 湍流结构的生成演化与作用机理重大研究计划年度交流会, 2023 年 1 月, 宁波.

“Perturbation Analysis and Perturbative Control of Buoyancy-Driven Flows”, 2023 年 1 月, 香港理工大学, 香港.

“Perturbation analysis of baroclinic torque in low-Mach-number flows”, 2022 年 11 月, 中南大学, 长沙.

学术兼职

审稿人: International Journal of Heat and Mass Transfer, Acta Mechanica Sinica, Physics of Fluids, Journal of Turbulence, European Journal of Mechanics-B/Fluids, Scientific Reports, Boundary Value Problems, Acta Aerodynamica Sinica, Chinese Journal of Theoretical and Applied Mechanics
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